

FINAL PROJECT

Phase 1: Data Architecture and Synthesis

Machine Learning Final Project

Welcome to the first stage of your capstone journey. Before you can build world-class models, you must act as a **Data Architect**. This phase focuses on the "Three S's": **Sourcing, Scrutiny, and Synthesis**. You aren't just finding data; you are constructing a high-fidelity foundation for a future "Blue Ocean" market solution.

I. The Research Objective

Identify a complex, real-world research topic that requires a multi-dimensional perspective. Your topic should be broad enough to benefit from diverse data sources (e.g., *Predicting Urban Economic Resilience* or *Personalized Healthcare Outcomes in Aging Populations*).

II. Data Acquisition (The Hunt)

You are required to find **five (5) unique datasets**. To ensure diversity and robust testing, each dataset **must** come from a different online source.

- **Examples:** Kaggle, UCI Machine Learning Repository, Google Dataset Search, AWS Open Data, World Bank, or specific API scrapes (Twitter/X, Reddit).
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III. Exploratory Data Analysis (EDA)

Using **Google Colab**, perform a rigorous audit of each dataset. For **each** of the five sources, your notebook must clearly document:

- **Dimensionality:** Number of samples (rows) and features (columns).
 - **Data Quality:** Identification of missing values and null counts.
 - **Outlier Detection:** Use box plots or Z-scores to identify "noise" versus "signals."
 - **Descriptive Statistics:** Mean, median, standard deviation, and distribution shapes for key numerical variables.
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IV. Feature Selection & Integration Strategy

Once you have audited your data, you must narrow your focus.

1. **The Top 10:** From the pool of all features across all five datasets, identify the **10 most impactful features** you intend to use for your final model. Justify why these ten are the "signal" in the noise.
 2. **The Merge Plan:** In your Colab notebook, outline the technical logic for merging these datasets.
 - Which keys (e.g., timestamps, ISO country codes, User IDs) will you use for joins?
 - Will you use an inner, outer, or left join? Explain the potential impact on your sample size.
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V. The Synthetic "Gap" Component

Real-world data is rarely perfect. Identify **one crucial feature** that is missing from your gathered datasets but would significantly improve your model's predictive power.

- **The Task:** Propose a method to generate **synthetic data** to fill this gap.
 - **The Methods:** Indicate whether you will use:
 - **Monte Carlo Simulations:** To model probability distributions based on historical variance.
 - **Generative Adversarial Networks (GANs):** To create high-fidelity synthetic samples that mimic the statistical properties of your existing data.
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Phase 1 Deliverables (For Trifold/Submission)

1. Colab Notebook: A clean, commented Python notebook showing the EDA for all 5 sources. **2. Feature Matrix:** A table listing your Top 10 features and their original source. **3. Integration Flowchart:** A visual representation of how the 5 sources merge into one master dataframe. **4. Synthetic Proposal:** A brief technical write-up on your missing feature and your chosen generation method (Monte Carlo vs. GANs).

Looking Ahead

- **Phase 2:** Feature Engineering, Model Selection (XGBoost, CNN, RNN, LSTM, Transformers, ...), Hyperparameter Tuning, and Deployment.
- **Phase 3:** The "Blue Ocean" Strategy—explaining how your model provides a unique, uncontested solution to a market problem.